

1. Expansion & Special Products

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Key Algebraic Identities

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)^2 = a^2 - 2ab + b^2$
- Difference of Two Squares: $a^2 - b^2 = (a - b)(a + b)$
- $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

Expanding Polynomials

Use the distributive law for multiplying expressions:

$$(2x + 3)(3x - 4) = 6x^2 - 8x + 9x - 12 = 6x^2 + x - 12.$$

2. Factorization Techniques

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Factorization by Grouping

For 4-term expressions: $ax + ay + bx + by = a(x + y) + b(x + y) = (a + b)(x + y)$.

Simplifying Algebraic Fractions

To simplify algebraic fractions, factorize numerator and denominator completely, then cancel common factors. Never cancel individual terms across addition or subtraction signs.